

Domácí úkol 6.2

Wednesday 25 February 2026

17:56

$$f\begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 2 \\ 3 \end{pmatrix}, \quad f\begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ -2 \\ -2 \end{pmatrix}, \quad \text{Necht } B = \left(\begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right) \text{ báze } V \leq \mathbb{R}^4.$$

$$\text{Potom } [f]_B^B = \left(\left[\begin{pmatrix} 4 \\ 1 \\ 2 \\ 3 \end{pmatrix} \right]_B \mid \left[\begin{pmatrix} 2 \\ 2 \\ -2 \\ -2 \end{pmatrix} \right]_B \right) = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$$

$\uparrow \quad \uparrow$
 $v_1 + 2v_2 \quad 2v_1 - 2v_2$

$$\text{vl. c } [f]_B^B \rightarrow \det \begin{pmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{pmatrix} = (\lambda-2)(\lambda+3)$$

$$[M_2]_B = \ker \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\} \rightarrow M_2 = \text{span} \left\{ 2 \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right\} = \text{span} \left\{ \begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix} \right\}$$

$$[M_3]_B = \ker \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 1 \\ -2 \end{pmatrix} \right\} \rightarrow M_3 = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right\} = \text{span} \left\{ \begin{pmatrix} 0 \\ 1 \\ -2 \\ -3 \end{pmatrix} \right\}$$

$$\text{Necht } C = \left(\begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -2 \\ -3 \end{pmatrix} \right), \quad \text{Potom } [f]_C^C = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}. \quad \begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix} = 1 \cdot \begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix} - 1 \cdot \begin{pmatrix} 0 \\ 1 \\ -2 \\ -3 \end{pmatrix}, \quad \text{tedy}$$

$$\begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix}_C = (1, -1) \rightarrow f^n \begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix} = f^n \left(1 \cdot \begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix} - 1 \cdot \begin{pmatrix} 0 \\ 1 \\ -2 \\ -3 \end{pmatrix} \right) = f^n \begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix} - f^n \begin{pmatrix} 0 \\ 1 \\ -2 \\ -3 \end{pmatrix} =$$

$$2^n \begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix} - (-3)^n \begin{pmatrix} 0 \\ 1 \\ -2 \\ -3 \end{pmatrix}$$